Unit: 6 Unit 2: Exponents Radicals
Topic: Radicals

Name: ____________
Date: ________

Homework (Jalapeno)

Q1. The ancient Egyptians used $\sqrt{16}$ to represent the side length of a square with an area of 16 square units. What is the value of $\sqrt{16}$?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10

Q2. The Babylonians approximated $\sqrt{8}$ as $2\frac{3}{4}$. What is the simplified form of $\sqrt{8}$?
(A) $2\sqrt{2}$
(B) $\sqrt{4} \cdot \sqrt{2}$
(C) $2\frac{3}{4}$
(D) $\sqrt{16}$
(E) $\sqrt{4}$

Q3. The ancient Greeks used the concept of $\sqrt{2}$ to study the ratio of the diagonal to the side length of a square. What is the value of $\sqrt{4} \cdot \sqrt{2}$?
(A) $2\sqrt{2}$
(B) $\sqrt{8}$
(C) 4
(D) 2
(E) $\sqrt{2}$

Q4. The ancient Chinese used $\sqrt[3]{27}$ to represent the edge length of a cube with a volume of 27 cubic units. What is the value of $\sqrt[3]{27}$?
(A) 2
(B) 3
(C) 6
(D) 9
(E) 12

Q5. The Mayans used $\sqrt{25}$ to calculate the area of a square with a side length of 5 units. What is the simplified form of $\sqrt{25}$?
(A) $\sqrt{5}$
(B) $\sqrt{25}$
(C) 5
(D) 10
(E) 25

Q6. The ancient Indians used $\sqrt[3]{64}$ to represent the edge length of a cube with a volume of 64 cubic units. What is the value of $\sqrt[3]{64}$?
(A) 2
(B) 3
(C) 4
(D) 6
(E) 8

Q7. The ancient Egyptians used $\sqrt{36}$ to calculate the area of a square with a side length of 6 units. What is the simplified form of $\sqrt{36}$?
(A) $\sqrt{6}$
(B) $\sqrt{36}$
(C) 6
(D) 12
(E) 36

Q8. The Babylonians used $\sqrt{9}$ to represent the side length of a square with an area of 9 square units. What is the value of $\sqrt{9}$?
(A) 2
(B) 3
(C) 6
(D) 9
(E) 12

Open Ended Questions (FRQ)

Q9. The ancient Greeks used the concept of $\sqrt{x}$ to study the ratio of the diagonal to the side length of a square. If $\sqrt{x} = 5$, what is the value of x? Show your work and explain your reasoning.

Q10. The Mayans used $\sqrt[3]{y}$ to calculate the edge length of a cube with a volume of y cubic units. If $\sqrt[3]{y} = 3$, what is the value of y? Show your work and explain your reasoning.

Chef's Tips

- Tip 1: Encourage students to use historical and ancient contexts to understand the concept of radicals.
- Tip 2: Remind students to simplify their answers and show their work.
- Tip 3: Emphasize the importance of understanding the concept of radicals in real-world applications.